

ITA0303 - mobile computing:

CO4 - AT2 : case Study.

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16) Analysis of the ios application:-

The delays and high battery consumption can be caused by the interaction between the ios runtime, OS services, application execution and Power management.

1) ios Runtime environment:-

- * ios uses ARC for memory management.

- * Intensive video processing can consume large amounts of CPU/GPU and memory.

- * If objects such as video buffers are retained longer than necessary,

memory pressure can increase and reduce performance.

2.) Application Execution:-

- * continuous video processing keeps the CPU/GPU active for long periods.
- * Heavy processing on the main thread can block UI operations, causing delayed touches, scrolling, and screen updates.
- * Poorly optimized loops or unnecessary background tasks can further increase CPU usage.

3.) operating - System Service:-

- * Notifications, networking, location services, and other system services compete for CPU, memory and I/O resources.
- * iOS prioritizes interactive tasks, but excessive application workloads

can still cause scheduling delays.

- * Background execution is also controlled by iOS to protect system responsiveness and battery life.

4.) Power management :-

- * Continuous CPU/GPU activity increases power consumption and device temperature.

- * iOS may reduce processor performance through thermal throttling when the device becomes hot.

- * This creates a cycle: heavy processing → high power/heat → throttling → slower processing.

5) Potential Architectural causes :-

- * Video Processing Performed Synchronously on the main thread.

- * Continuous Processing Instead of Processing only when required.

- * Inefficient memory/buffer management.
- * Excessive background activity and frequent notification.
- * Failure to use efficient hardware-accelerated video processing APIs.

Conclusion :-

The main cause is resource contention between video - processing, UI tasks, OS services, and Power management.